

Designing and Conducting a Continual Reassessment Method trial using Dose Transition Pathways

Christina Yap, Daniel Slade, Kristian Brock and Yi Pan

2019-03-30

Design a CRM trial with DTP

Building on the R `dfcrm` package [1], the `dtprcm` package provides the Dose Transition Pathways (DTP) to project in advance the doses recommended by a model-based design for subsequent patients (stay, escalate, de-escalate or stop early) using all the accumulated toxicity information [2]. DTP can be used as a design calibration tool and an operational tool. It can be presented as a table or as a flow diagram. The `dtprcm` package also provides the modified Continual Reassessment Method (CRM) [3] and Time to event CRM (TITE-CRM) [4] with added practical considerations to allow stopping early when:

- there is sufficient evidence that the lowest dose is too toxic and/or
- there is a sufficient number of patients dosed at the maximum tolerated dose

Here, we will illustrate the use of the DTP as

1. a design calibration tool
2. an analysis and operational tool

when designing a CRM, using the Viola trial [2,5].

Input

We first need to input the parameters required for design considerations for CRM as displayed in Figure 1 in Yap et al [2] which were grouped under three broad categories of Clinical Parameters, Model Specification parameters and Practical Considerations.

```
library(dtprcm)
set.seed(1000)
```

1. Clinical Parameters

```
number.doses= 7

start.dose.level = 3

max.sample.size = 21

target.DLT = 0.2

cohort.size = 3
```

2. Model Parameters

A one-stage one-parameter Bayesian empiric model is used where the slope parameter is assumed to be normally distributed with mean 0 with a specified prior variance.

```
prior.DLT = c(0.03, 0.07, 0.12, 0.20, 0.30, 0.40, 0.52)
prior.var = 0.75
```

3. Practical Considerations

Practical considerations for Viola include:

- No skipping of doses in escalation (default option)
- Allow skipping of doses in de-escalation.
- No dose escalation if the overall observed rate of DLT at the current dose level is above the target DLT rate (default option)

```
no_skip_esc = TRUE
no_skip_deesc = FALSE
global_coherent_esc = TRUE
```

- Stop the trial early for toxicity if there is strong evidence that the lowest dose is too toxic based on a Bayesian safety early criterion:

$p(\text{true DLT rate at lowest dose} > \text{target DLT rate} + x\% \mid \text{current observed data and any relevant prior information}) > y$

```
y = 0.99
x = 0.05
tox_lim = target.DLT + x
```

- Stop the trial early if there are 12 patients at the recommended MTD.

```
n.mtd= 12
```

Stopping early criterion (for safety and sufficient number of patients at MTD)

```
stop_func <- function(x) {  
  y = stop_for_excess_toxicity_empiric(x, tox_lim = tox_lim, prob_cert = y, dose = 1, nsamps =  
10000)  
  if(y$stop){  
    x <- y  
  } else {x = stop_for_consensus_reached(x, req_at_mtd = n.mtd)}  
}
```

Simulations: assess operating characteristics of the proposed CRM design

Let's examine three scenarios, S1, S2 and S3 with true DLT rates at each dose as defined below. S1 and S2 presents scenarios where the true MTD are at d(1) and d(3) respectively. S3 presents a setting whereby all doses are too toxic.

```
S1=c(0.03, 0.05, 0.07, 0.20, 0.36, 0.45, 0.55)  
S2=c(0.005, 0.01, 0.03, 0.05, 0.07, 0.20, 0.36)  
S3=c(0.45, 0.55, 0.65, 0.7, 0.75, 0.80, 0.85)
```

	d(-2)	d(-1)	d(0)	d(1)	d(2)	d(3)	d(4)
S1	0.030	0.05	0.07	0.20	0.36	0.45	0.55
S2	0.005	0.01	0.03	0.05	0.07	0.20	0.36
S3	0.450	0.55	0.65	0.70	0.75	0.80	0.85

```
No.of.simulations = 5
```

Scenario 1: True MTD at d(1),

```
sim1_1 <- applied_crm_sim(true_tox = S1, prior = prior.DLT, target = target.DLT,  
max_sample_size=max.sample.size, first_dose=start.dose.level, num_sims = No.of.simulations,  
cohort_size = cohort.size, dose_func = applied_crm, scale = sqrt(prior.var), stop_func =  
stop_func)
```

Extracting outputs on probability of selecting each dose as the MTD and the number of subjects allocated to each dose

```
out1_1<- rbind.data.frame("S1 True Tox" = sim1_1$summary$true_tox,  
"S1 Prob Select"= sim1_1$summary$mtd,  
"S1 No.Subjects"= sim1_1$summary$doses_given)  
dimnames(out1_1)[[2]]<-paste("d(", seq(-2,4,by=1),")", sep="")  
#out1_1  
#sim1_1$summary$prob_stop[2]
```

Repeat the procedure for Scenarios 2 and 3.

```
sim1_3$summary$prob_stop[2]  
#> [1] NA  
# to check if we can extract out probability of stopping early due to toxicity  
  
out<- cbind.data.frame(out1_3, Stop.Tox= c(NA, NA, sim1_1$summary$prob_stop[2], NA, NA,  
sim1_2$summary$prob_stop[2], NA, NA, sim1_3$summary$prob_stop[2], NA), Stop.nMTD= c(NA, NA,  
sim1_1$summary$prob_stop[1], NA, NA, sim1_2$summary$prob_stop[1], NA, NA,  
sim1_3$summary$prob_stop[1], NA) )
```

Simulation Results

The table below presents the simulation results for the selected scenarios, with true DLT rates, probability of selecting a dose as the MTD and the number of subjects allocated to each dose. It also presents the probability of stopping early due to toxicity or having sufficient number of patients at the MTD.

	d(-2)	d(-1)	d(0)	d(1)	d(2)	d(3)	d(4)	Stop.Tox	Stop.nMTD
prior	0.030	0.07	0.12	0.20	0.30	0.40	0.52	NA	NA
S1 True Tox	0.030	0.05	0.07	0.20	0.36	0.45	0.55	NA	NA
S1 Prob Select	0.000	0.00	0.00	0.60	0.40	0.00	0.00	NA	0.4
S1 No.Subjects	0.000	1.80	5.40	8.40	5.40	0.00	0.00	NA	NA
S2 True Tox	0.005	0.01	0.03	0.05	0.07	0.20	0.36	NA	NA
S2 Prob Select	0.000	0.00	0.00	0.00	0.00	0.60	0.40	NA	0.2
S2 No.Subjects	0.000	0.00	3.00	3.60	4.80	6.00	3.60	NA	NA
S3 True Tox	0.450	0.55	0.65	0.70	0.75	0.80	0.85	NA	NA
S3 Prob Select	1.000	0.00	0.00	0.00	0.00	0.00	0.00	NA	1.0
S3 No.Subjects	12.000	3.00	3.00	0.00	0.00	0.00	0.00	NA	NA

Calculate DTP for cohorts 1-4 in Viola

Using the specific parameters above, we can produce the DTP for the initial 4 cohorts.

```
start.dose.level<-3    #(eg. 1,2,3 etc)

viola_dtps <- calculate_dtps(next_dose = start.dose.level, cohort_sizes = c(cohort.size,
cohort.size, cohort.size), dose_func = applied_crm, prior = prior.DLT, target = target.DLT,
stop_func = stop_func, scale = sqrt(prior.var),
no_skip_esc = no_skip_esc, no_skip_deesc = no_skip_deesc,
global_coherent_esc = global_coherent_esc)
```

In Viola, there are 7 dose levels where the starting dose level is 3. For display purposes, we would use the dose-labels as used in Viola, which range from d(-2) to d(4), with starting dose at d(0).

```
# Using dose labels
viola_dtps[seq(1, ncol(viola_dtps), by=2)] <- viola_dtps[seq(1, ncol(viola_dtps), by=2)] -
start.dose.level
dimnames(viola_dtps)[[2]]<-c('C1 Dose', 'C1 No.DLT', 'C2 Dose', 'C2 No.DLT','C3 Dose', 'C3
No.DLT', 'C4 Dose')
```

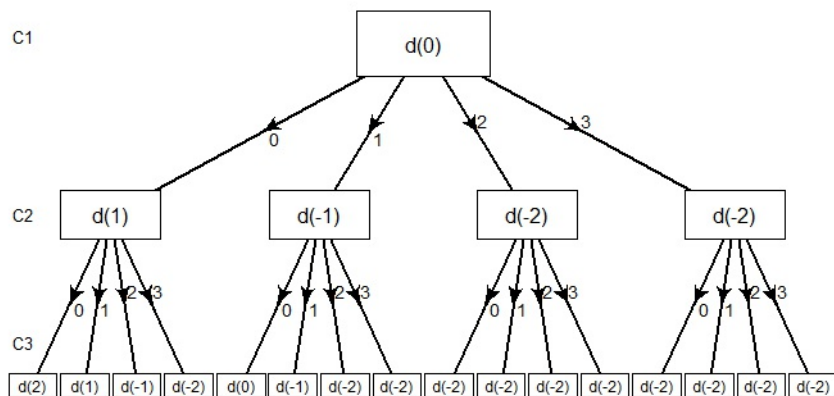
Displaying DTP in a table format

C1 Dose	C1 No.DLT	C2 Dose	C2 No.DLT	C3 Dose	C3 No.DLT	C4 Dose
0	0	1	0	2	0	3
0	0	1	0	2	1	2
0	0	1	0	2	2	1
0	0	1	0	2	3	0
0	0	1	1	1	0	2
0	0	1	1	1	1	0
0	0	1	1	1	2	-1
0	0	1	1	1	3	-2
0	0	1	2	-1	0	0
0	0	1	2	-1	1	-1
0	0	1	2	-1	2	-2
0	0	1	2	-1	3	-2
0	0	1	3	-2	0	-1
0	0	1	3	-2	1	-2
0	0	1	3	-2	2	-2
0	0	1	3	-2	3	-2
0	1	-1	0	0	0	1
0	1	-1	0	0	1	0
0	1	-1	0	0	2	-1
0	1	-1	0	0	3	-2
0	1	-1	1	-1	0	-1
0	1	-1	1	-1	1	-2
0	1	-1	1	-1	2	-2
0	1	-1	1	-1	3	-2
0	1	-1	2	-2	0	-2
0	1	-1	2	-2	1	-2
0	1	-1	2	-2	2	-2
0	1	-1	2	-2	3	-2
0	1	-1	3	-2	0	-2
0	1	-1	3	-2	1	-2
0	1	-1	3	-2	2	-2
0	1	-1	3	-2	3	-2
0	2	-2	0	-2	0	-1
0	2	-2	0	-2	1	-2
0	2	-2	0	-2	2	-2
0	2	-2	0	-2	3	-2
0	2	-2	1	-2	0	-2
0	2	-2	1	-2	1	-2
0	2	-2	1	-2	2	-2
0	2	-2	1	-2	3	-2
0	2	-2	2	-2	0	-2

0	2	-2	2	-2	1	-2
0	2	-2	2	-2	2	-2
0	2	-2	2	-2	3	-2
0	2	-2	3	-2	0	-2
0	2	-2	3	-2	1	-2
0	2	-2	3	-2	2	-2
0	2	-2	3	-2	3	NA
0	3	-2	0	-2	0	-2
0	3	-2	0	-2	1	-2
0	3	-2	0	-2	2	-2
0	3	-2	0	-2	3	-2
0	3	-2	1	-2	0	-2
0	3	-2	1	-2	1	-2
0	3	-2	1	-2	2	-2
0	3	-2	1	-2	3	-2
0	3	-2	2	-2	0	-2
0	3	-2	2	-2	1	-2
0	3	-2	2	-2	2	-2
0	3	-2	2	-2	3	NA
0	3	-2	3	-2	0	-2
0	3	-2	3	-2	1	-2
0	3	-2	3	-2	2	NA
0	3	-2	3	-2	3	NA

Displaying DTP as a flow diagram for Cohorts 1-3

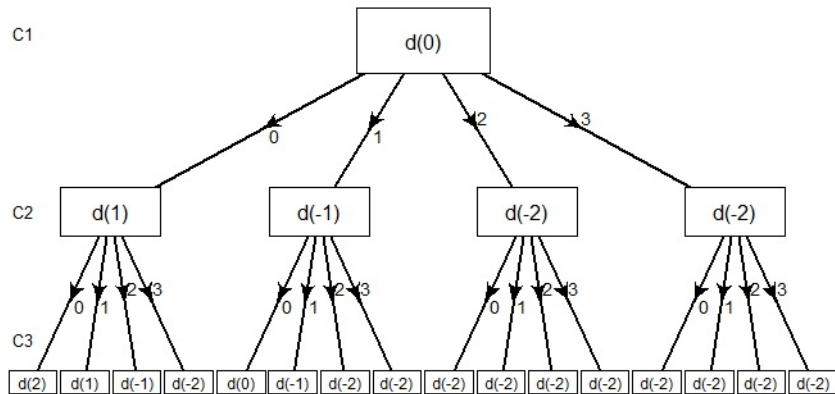
```
dtflow(viola_dtfs)
```



One can utilise the DTP to discuss with the investigators on the working of the proposed CRM design for the initial cohorts, tailoring the model to the specific requirements of the trial. In particular, it would be useful to check that the investigators are in agreement on when a trial should be stopped early due to toxicity if excessive DLTs are observed in the initial cohorts. The design may then be re-calibrated to provide DTP that the investigators are happy to adopt in practice. with. The final recommended model should be one that has promising operating characteristics with dose pathways that are acceptable for the trial [2,5,6].

In the case for Viola, the investigators were keen to only stop the trial early if they saw 2 or 3 DLTs out of the first 3 patients at the lowest dose, d(-2), which would indicate that the doses are too toxic. This guided the choice of γ to be around 0.72.

```
dtflow(viola_dtfs)
```



Trial Conduct

No DLTs were observed in the first 2 cohorts who were dosed at dose levels 3 and 4.

```
DLT.outcomes<-c(0,0,0,0,0,0)
dose.level = c(3,3,3,4,4,4)
```

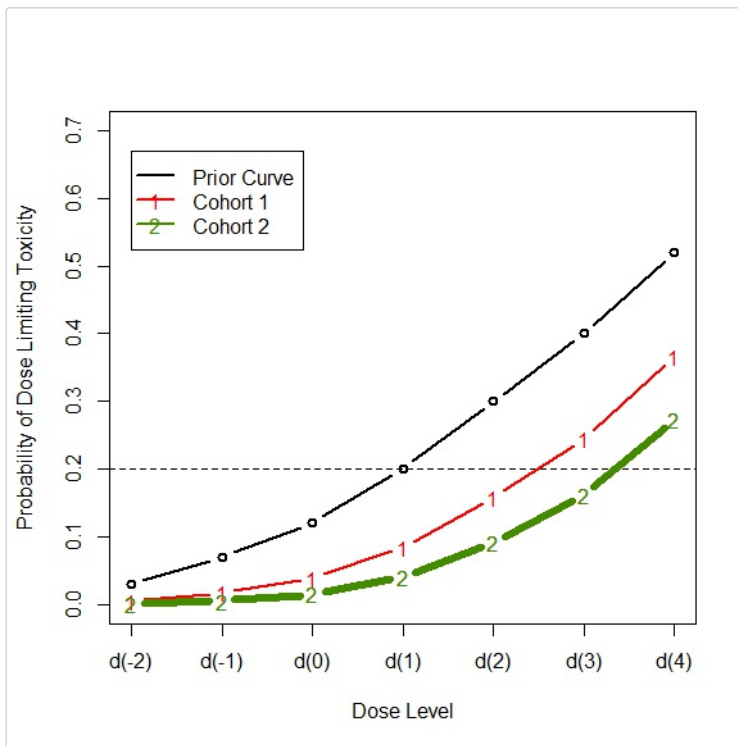
Model Updates of DLT rates with accumulated information

```
C2 <- applied_crm(prior=prior.DLT, target = target.DLT, tox = DLT.outcomes, level = dose.level,
  stop_func = stop_func,
  no_skip_esc = no_skip_esc,
  no_skip_deesc = no_skip_deesc,
  global_coherent_esc = global_coherent_esc,
  scale = sqrt(prior.var))
results<-cbind.data.frame(Dose=paste0("d(", dose.labels, ")"), summary_crm(C2)
[2:ncol(summary_crm(C2))])
knitr::kable(results)
```

Dose	Prior Prob(DLT)	Number of Evaluable Patients	Number of DLTs	Posterior Prob(DLT)
d(-2)	0.03	0	0	0.001 (0, 0.079)
d(-1)	0.07	0	0	0.005 (0, 0.146)
d(0)	0.12	3	0	0.015 (0, 0.216)
d(1)	0.20	3	0	0.041 (0, 0.312)
d(2)	0.30	0	0	0.091 (0.001, 0.419)
d(3)	0.40	0	0	0.162 (0.007, 0.515)
d(4)	0.52	0	0	0.273 (0.028, 0.623)

Update of Dose Toxicity Curves

```
plot_crm(C2, dose_labels = paste0("d(",dose.labels,")"), cohort_sizes = c(3,3), dose_func
=applied_crm,
  no_skip_esc = no_skip_esc,
  no_skip_deesc = no_skip_deesc,
  global_coherent_esc = global_coherent_esc,
  scale = sqrt(prior.var),
  ylim = c(0, 0.7), lwd = 2, cex.axis = 1.5, cex.lab = 1.4, cex = 1, cohort.last = T
)
```



source code changed for xlab to be "Dose"

Update Dose Transtion Pathways for Subsequent Cohorts

DTP for subsequent cohorts can be updated regularly. For instance, we can update after Cohorts 1 and 2, giving the DTP for Cohorts 3-5.

```

dtp_c3 <- calculate_dtps(next_dose = 5, cohort_sizes = c(3, 3, 3),
  prev_tox = DLT.outcomes, prev_dose = dose.level,
  prior = prior.DLT, target = target.DLT,
  no_skip_esc = no_skip_esc,
  no_skip_deesc = no_skip_deesc,
  global_coherent_esc = global_coherent_esc,
  stop_func = stop_func,
  scale = sqrt(prior.var))

# Using dose labels
dtp_c3[seq(1, ncol(dtp_c3), by=2)] <- dtp_c3[seq(1, ncol(dtp_c3), by=2)] - start.dose.level
dimnames(dtp_c3)[[2]]<-c('C3 Dose', 'C3 No.DLT', 'C4 Dose', 'C4 No.DLT', 'C5 Dose', 'C5 No.DLT',
'C6 Dose')

```

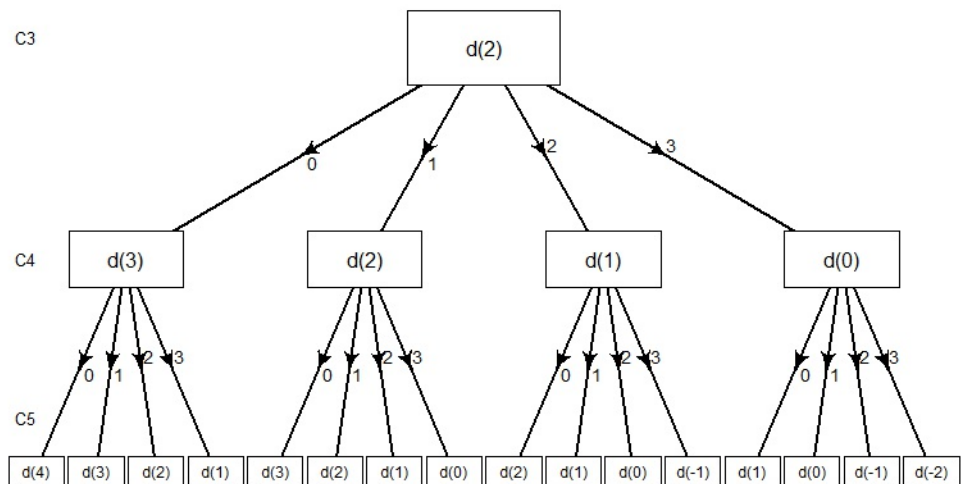
Updated DTP table for Cohorts 3 to 5

C3 Dose	C3 No.DLT	C4 Dose	C4 No.DLT	C5 Dose	C5 No.DLT	C6 Dose
2	0	3	0	4	0	4
2	0	3	0	4	1	4
2	0	3	0	4	2	3
2	0	3	0	4	3	2
2	0	3	1	3	0	3
2	0	3	1	3	1	3
2	0	3	1	3	2	2
2	0	3	1	3	3	1
2	0	3	2	2	0	3
2	0	3	2	2	1	2
2	0	3	2	2	2	1
2	0	3	2	2	3	0
2	0	3	3	1	0	2
2	0	3	3	1	1	1
2	0	3	3	1	2	0
2	0	3	3	1	3	-1
2	1	2	0	3	0	3
2	1	2	0	3	1	2
2	1	2	0	3	2	2
2	1	2	0	3	3	1
2	1	2	1	2	0	2

2	1	2	1	2	1	1
2	1	2	1	2	2	1
2	1	2	1	2	3	0
2	1	2	2	1	0	1
2	1	2	2	1	1	1
2	1	2	2	1	2	0
2	1	2	2	1	3	-1
2	1	2	3	0	0	1
2	1	2	3	0	1	0
2	1	2	3	0	2	-1
2	1	2	3	0	3	-2
2	2	1	0	2	0	2
2	2	1	0	2	1	1
2	2	1	0	2	2	1
2	2	1	0	2	3	0
2	2	1	1	1	0	1
2	2	1	1	1	1	0
2	2	1	1	1	2	0
2	2	1	1	1	3	-1
2	2	1	2	0	0	0
2	2	1	2	0	1	-1
2	2	1	2	0	2	-1
2	2	1	2	0	3	-2
2	2	1	3	-1	0	-1
2	2	1	3	-1	1	-1
2	2	1	3	-1	2	-2
2	2	1	3	-1	3	-2
2	3	0	0	1	0	1
2	3	0	0	1	1	0
2	3	0	0	1	2	0
2	3	0	0	1	3	-1
2	3	0	1	0	0	0
2	3	0	1	0	1	-1
2	3	0	1	0	2	-1
2	3	0	1	0	3	-2
2	3	0	2	-1	0	-1
2	3	0	2	-1	1	-2
2	3	0	2	-1	2	-2
2	3	0	2	-1	3	-2
2	3	0	3	-2	0	-2
2	3	0	3	-2	1	-2
2	3	0	3	-2	2	-2
2	3	0	3	-2	3	-2

DTP flow diagram for Cohorts 3 to 5

```
dtpflow(dtp_c3, cohort.labels = c('C3', 'C4', 'C5'))
```



References

- [1] Ken Cheung (2013). dcfm: Dose-finding by the continual reassessment method. R package version 0.2-2. <https://CRAN.R-project.org/package=dfcm>
- [2] Yap, C., Billingham, L. J., Cheung, Y. K., Craddock, C., & O'Quigley, J. (2017). Dose Transition Pathways: The missing link between complex dose-finding designs and simple decision-making. *Clinical Cancer Research*, 23(24), 7440-7447
- [3] O'Quigley, J. O., Pepe, M., and Fisher, L. (1990). Continual reassessment method: A practical design for phase I clinical trials in cancer. *Biometrics* 46:33-48.
- [4] Cheung YK, Chappell R. Sequential designs for phase I clinical trials with late onset toxicities. *Biometrics* 2000;56:1177-82.
- [5] Craddock C, Slade D, Santo C De, Wheat R, Ferguson P, Hodgkinson A, Brock K, Cavenagh J, Ingram W, Dennis M, Malladi R, Siddique S, Mussai F and Yap C (2018) Combination lenalidomide and azacitidine: a novel salvage therapy in patients who relapse after allogeneic stem cell transplantation for acute myeloid leukemia. *Journal of Clinical Oncology* (2019), JCO-18.
- [6] Wheeler, G.M., Mander, A.P., Bedding, A., Brock, K., Cornelius, V., Grieve, A.P., Jaki, T., Love, S.B., Weir, C.J., Yap, C. and Bond, S.J., 2019. How to design a dose-finding study using the continual reassessment method. *BMC medical research methodology*, 19(1), p.18.